

Daniel Kim

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EDUCATION

Rice University

4.0/4.0

Bachelor of Science in Electrical and Computer Engineering and Physics

August 2024 – May 2028

- **Scholarships:** Max Roy Scholarship (1 recipient, full tuition, merit-based)
Los Alamos Lab. Gold Scholarship (3 recipients, \$20,000, merit, leadership, service-based)
- **Coursework:** Data Structures and Algorithms, Digital Logic, Signals and Systems, Random Signals, Analog Circuits, Linear Algebra

SKILLS

Software: C++ (CUDA, OpenGL, OpenCV), Python (PyTorch, NumPy), Dart, TypeScript, MATLAB (DSP Builder, HDL Coder), Java, Swift

Frameworks: ROS, Git, Linux/Unix, Google Cloud, Jupyter, Docker, Intel Quartus/Platform Designer, AMD Vivado

Hardware: RTL Design (Verilog), Altium, KiCAD, Autodesk Fusion360, Arduino, PCB design, Soldering/Reflow, 3D Printing

WORK AND RESEARCH EXPERIENCE

Rice Computational Imaging Lab

Houston, TX

Undergraduate Research Assistant

August 2025 – Present

- Engineering GPU-accelerated machine learning algorithms (CUDA/C++) for large-scale implicit neural representations.
- Accelerating and scaling deep learning pipelines to reduce runtime and memory footprint on high-resolution datasets.

NASA Kennedy Space Center

Cape Canaveral, FL

Robotics Software Engineering Intern

May 2025 – August 2025

- Developed a ROS-based tracking system, implementing computer vision for real-time rocket motion tracking and segmentation.
- Optimized control system performance by developing a neural-network-based PID controller, reducing tracking error by 94%.

Los Alamos National Laboratory

Los Alamos, NM

Electrical Engineer R&D Intern

June 2023 – August 2024

- Streamlined RF field stabilization by developing an FPGA-based simulator, eliminating need for ~\$350K in equipment.
- Replaced manual PID tuning with robust self-tuning control algorithms for precise temperature control in particle accelerators.

Joint Science and Technology Institute West

Albuquerque, NM

Computer Science Researcher

June 2023

- Achieved a 92% runtime improvement in MPAS Ocean through tracer advection modeling in collaboration with student research group.
- Integrated tensor algebra with heat flow physics to create GPU and TPU-based heat transfer models in different mediums.

PUBLICATIONS

- Vyas, K., Kayabasi, A., **Kim, D.**, Saragadam, V., Veeraraghavan, A., & Balakrishnan, G. (2025). *The Surprising Effectiveness of Noise Pretraining for Implicit Neural Representations*. Submitted to IEEE/CVF CVPR 2026.
- **Kim, D.**, & Iturregui, A. (2022). *Developing a Control Algorithm and Simulation for Thrust Vector Controlled Rockets*. New Mexico Journal of Science.
- **Kim, D.** (2024). *Engineered Underwater Vehicle for Ocean Litter Mapping*. New Mexico Journal of Science.

PROJECTS

app.studybeats.co: Interactive web-based productivity suite (Dart, Google Cloud)

November 2023 – August 2025

- Engineered and deployed a scalable, full-stack web application, with real-time data synchronization.
- Implemented user authentication, database optimization, and CI/CD pipelines, supporting 800+ concurrent users.

OceanAI: Underwater vehicle and computer vision for mapping ocean trash (C++, TypeScript, Python, KiCAD)

June 2022 – May 2023

- Developed autonomous underwater vehicle to capture subsea footage and process it with YOLOv5-based object detection,
- Implemented deep learning and SLAM-based mapping to discover 174,000+ pieces of ocean trash, winning Regeneron ISEF 3rd place.

OrbitSim: Orbital simulator desktop app (C++, OpenGL, ImGUI).

July 2023 – October 2023

- Simulated and visualized over 8000 active satellite trajectories and gravitational interactions using realistic physics models.
- Implemented a modular graphics pipeline with optimized memory usage, enabling smooth interactive 3D visualization and modeling.

NM Rocketry: Thrust vector-controlled model rocket (C++, PCB Design, Fusion360, Simulink)

November 2021 – May 2022

- Engineered a real-time flight stabilization system, implementing PID with Kalman filter to actively control rocket trajectory.
- Developed PCB hardware and custom I2C/SPI drivers to achieve low-latency, high-reliability sensor data collection.